

Initiating Search

November 10, 2025, 9:49 PM

• Search:

CAS Lexicon Search:

Fischer indole synthesis - Preferred Concept

Search Tasks

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Yield:	90-100%, 80-89%, 70-79%	
Document Type:	Journal	

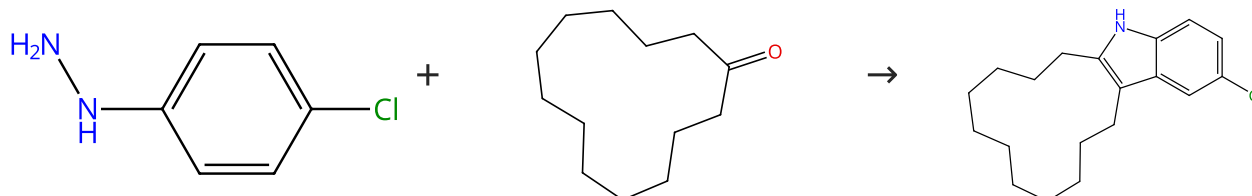
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 Reactions (50)[View in CAS SciFinder](#)

Scheme 1 (1 Reaction)

Steps: 1 Yield: 100%

 Suppliers (65) Suppliers (20)

31-467-CAS-18180575

Steps: 1 Yield: 100%

Fischer indole synthesis in the absence of a solvent

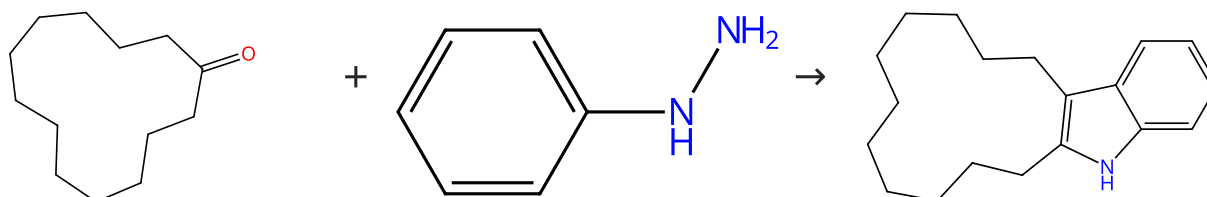
1.1 Reagents: *p*-Toluenesulfonic acid; 1 min, 250 °C

By: Matsumoto, Kiyoshi; et al

Heterocyclic Communications (2003), 9(1), 9-12.

Scheme 2 (1 Reaction)

Steps: 1 Yield: 100%

 Suppliers (20) Suppliers (64) Suppliers (2)

31-467-CAS-18210372

Steps: 1 Yield: 100%

Fischer indole synthesis in the absence of a solvent

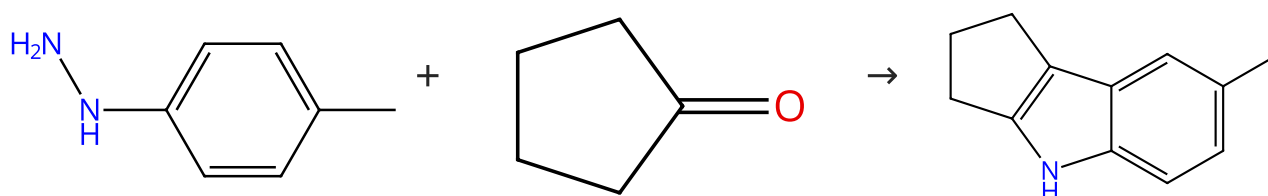
1.1 Reagents: Trichloroacetic acid; 5 min, 100 °C

By: Matsumoto, Kiyoshi; et al

Heterocyclic Communications (2003), 9(1), 9-12.

Scheme 3 (1 Reaction)

Steps: 1 Yield: 100%

 Suppliers (54) Suppliers (78) Suppliers (26)

31-467-CAS-18276855

Steps: 1 Yield: 100%

Fischer indole synthesis in the absence of a solvent

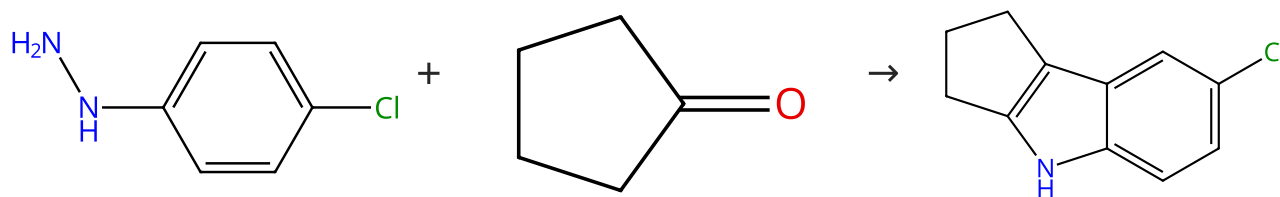
1.1 Reagents: *p*-Toluenesulfonic acid; 1 min, 250 °C

By: Matsumoto, Kiyoshi; et al

Heterocyclic Communications (2003), 9(1), 9-12.

Scheme 4 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (65)

Suppliers (78)

Suppliers (25)

31-467-CAS-14737063

Steps: 1 Yield: 100%

Fischer indole synthesis in the absence of a solvent

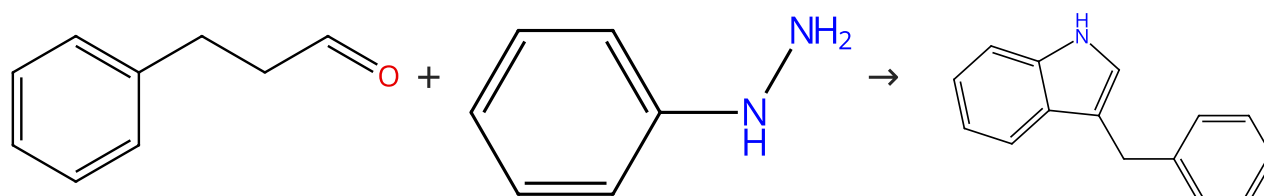
1.1 Reagents: *p*-Toluenesulfonic acid; 1 min, 250 °C

By: Matsumoto, Kiyoshi; et al

Heterocyclic Communications (2003), 9(1), 9-12.

Scheme 5 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (90)

Suppliers (64)

Suppliers (35)

31-467-CAS-3552516

Steps: 1 Yield: 100%

A rapid and reliable assay for regioselectivity using fluorescence spectroscopy

1.1 Catalysts: Sulfuric acid

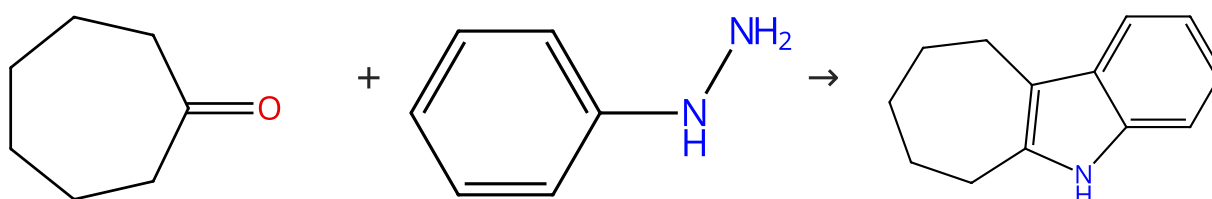
By: Angelovski, Goran; et al

Solvents: Tetrahydrofuran; 3 h, reflux; reflux → rt

Advanced Synthesis & Catalysis (2006), 348(10+11), 1193-1199.

Scheme 6 (1 Reaction)

Steps: 1 Yield: 100%



Suppliers (65)

Suppliers (64)

Suppliers (53)

31-467-CAS-18204493

Steps: 1 Yield: 100%

Fischer indole synthesis in the absence of a solvent

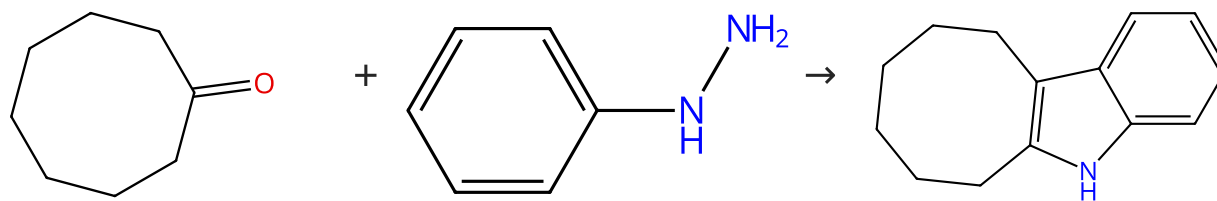
1.1 Reagents: Trichloroacetic acid; 5 min, 100 °C

By: Matsumoto, Kiyoshi; et al

Heterocyclic Communications (2003), 9(1), 9-12.

Scheme 7 (2 Reactions)

Steps: 1 Yield: 97-100%



Suppliers (80)

Suppliers (64)

Suppliers (49)

31-467-CAS-15489530

Steps: 1 Yield: 100%

1.1 **Catalysts:** Trifluoroacetic acid
Solvents: Acetic acid; 2 h, 90 °C

Experimental Protocols

A study on the synthesis of structural analogs of bis- indole alkaloid caulerpin: a step-by-step synthesis of a cyclic indole-tetramer

By: Talaz, Oktay; et al

Tetrahedron (2010), 66(10), 1902-1910.

31-467-CAS-9948257

Steps: 1 Yield: 97%

1.1 -
 1.2 **Reagents:** Acetic acid, Sulfuric acid

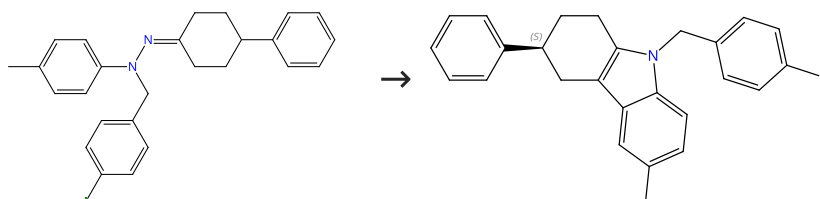
Indole as a Tool in Synthesis. Algorithmic Construction of a Family of Compounds with all Ring Sizes Ranging from 10 to 16

By: Sigaut, F.; et al

Tetrahedron (2000), 56(49), 9641-9646.

Scheme 8 (1 Reaction)

Steps: 1 Yield: 99%



Absolute stereochemistry shown

31-467-CAS-13071005

Steps: 1 Yield: 99%

1.1 **Catalysts:** Diindenyl[7,1-*d*:1',7'-*fg*][1,3,2]dioxaphosphocin, 3, 7-di-9-anthracenyl-10,11,12,13-tetrahydro-5-hydroxy-, 5-oxide, (11*aR*)-
Solvents: Benzene; 4 - 5 d, 30 °C

Experimental Protocols

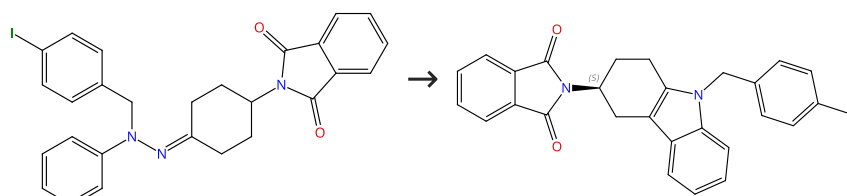
The Catalytic Asymmetric Fischer Indolization

By: Mueller, Steffen; et al

Journal of the American Chemical Society (2011), 133(46), 18534-18537.

Scheme 9 (1 Reaction)

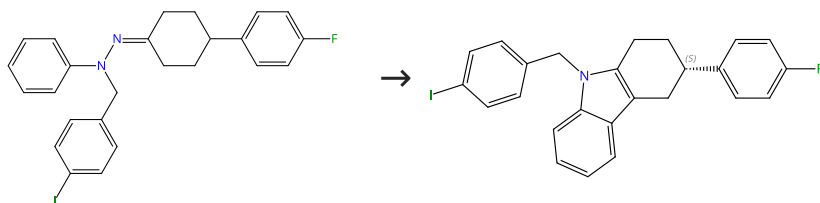
Steps: 1 Yield: 99%

Absolute stereochemistry shown,
Rotation (-)

31-467-CAS-14541841	Steps: 1 Yield: 99%	The Catalytic Asymmetric Fischer Indolization
1.1 Catalysts: Diindeno[7,1- <i>de</i> :1',7'- <i>fg</i>][1,3,2]dioxaphosphocin, 3, 7-di-9-anthracenyl-10,11,12,13-tetrahydro-5-hydroxy-, 5-oxide, (11a <i>R</i>)- Solvents: Benzene; 6 d, 30 °C		By: Mueller, Steffen; et al Journal of the American Chemical Society (2011), 133(46), 18534-18537.
Experimental Protocols		

Scheme 10 (1 Reaction)

Steps: 1 Yield: 99%

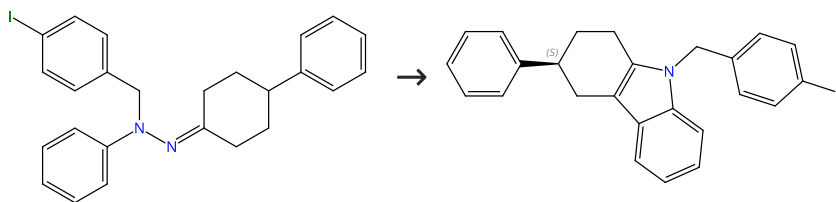


Absolute stereochemistry shown

31-467-CAS-14863506	Steps: 1 Yield: 99%	The Catalytic Asymmetric Fischer Indolization
1.1 Catalysts: Diindeno[7,1- <i>de</i> :1',7'- <i>fg</i>][1,3,2]dioxaphosphocin, 3, 7-di-9-anthracenyl-10,11,12,13-tetrahydro-5-hydroxy-, 5-oxide, (11a <i>R</i>)- Solvents: Benzene; 4 - 5 d, 30 °C		By: Mueller, Steffen; et al Journal of the American Chemical Society (2011), 133(46), 18534-18537.
Experimental Protocols		

Scheme 11 (1 Reaction)

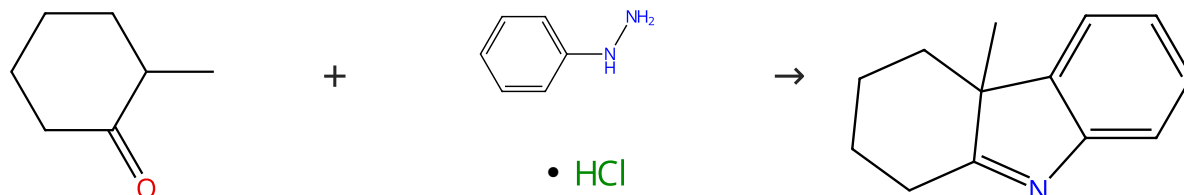
Steps: 1 Yield: 99%

Absolute stereochemistry shown,
Rotation (-)

31-467-CAS-10941961	Steps: 1 Yield: 99%	The Catalytic Asymmetric Fischer Indolization
1.1 Catalysts: Diindeno[7,1- <i>de</i> :1',7'- <i>fg</i>][1,3,2]dioxaphosphocin, 3, 7-di-9-anthracenyl-10,11,12,13-tetrahydro-5-hydroxy-, 5-oxide, (11a <i>R</i>)- Solvents: Benzene; 4 - 5 d, 30 °C		By: Mueller, Steffen; et al Journal of the American Chemical Society (2011), 133(46), 18534-18537.
Experimental Protocols		

Scheme 12 (1 Reaction)

Steps: 1 Yield: 99%



Suppliers (85)

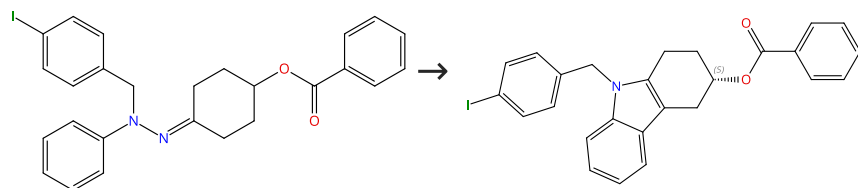
Suppliers (64)

Suppliers (9)

31-467-CAS-10886278	Steps: 1 Yield: 99%	Fischer Indole Synthesis in Low Melting Mixtures
1.1 Catalysts: (+)-Tartaric acid, Dimethylolurea; 0.5 h, 70 °C 1.2 Reagents: Water		By: Gore, Sangram; et al Organic Letters (2012), 14(17), 4568-4571.
Experimental Protocols		

Scheme 13 (1 Reaction)

Steps: 1 Yield: 99%



Absolute stereochemistry shown

31-467-CAS-12414017

Steps: 1 Yield: 99%

The Catalytic Asymmetric Fischer Indolization

1.1 **Catalysts:** Diindeno[7,1-*de*:1',7'-*fg*][1,3,2]dioxaphosphocin, 3, 7-di-9-anthracenyl-10,11,12,13-tetrahydro-5-hydroxy-, 5-oxide, (11a*R*)-

Solvents: Benzene; 4 - 5 d, 30 °C

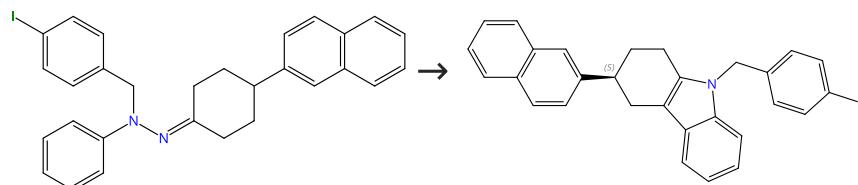
By: Mueller, Steffen; et al

Journal of the American Chemical Society (2011), 133(46), 18534-18537.

Experimental Protocols

Scheme 14 (1 Reaction)

Steps: 1 Yield: 99%



Absolute stereochemistry shown

31-467-CAS-1726187

Steps: 1 Yield: 99%

The Catalytic Asymmetric Fischer Indolization

1.1 **Catalysts:** Diindeno[7,1-*de*:1',7'-*fg*][1,3,2]dioxaphosphocin, 3, 7-di-9-anthracenyl-10,11,12,13-tetrahydro-5-hydroxy-, 5-oxide, (11a*R*)-

Solvents: Benzene; 4 - 5 d, 30 °C

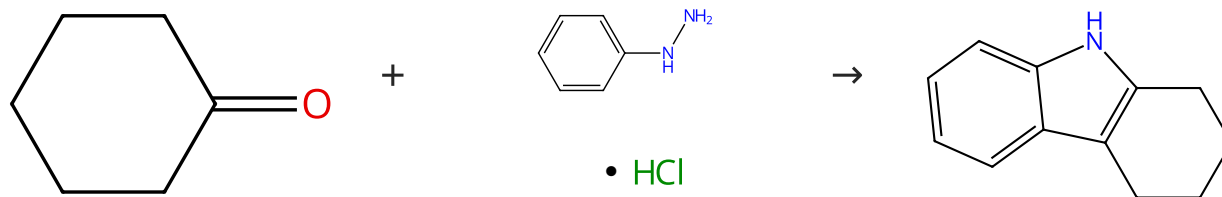
By: Mueller, Steffen; et al

Journal of the American Chemical Society (2011), 133(46), 18534-18537.

Experimental Protocols

Scheme 15 (3 Reactions)

Steps: 1 Yield: 96-99%



Suppliers (102)

Suppliers (64)

Suppliers (70)

31-467-CAS-11583316

Steps: 1 Yield: 99%

Mild, efficient Fischer indole synthesis using 2,4,6-trichloro-1,3,5-triazine (TCT)

1.1 **Catalysts:** Cyanuric chloride
Solvents: Ethanol; 2 h, 80 °C; 80 °C → rt

By: Siddalingamurthy, Eranna; et al

Tetrahedron Letters (2013), 54(41), 5591-5596.

31-467-CAS-8153

Steps: 1 Yield: 97%

Fischer Indole Synthesis in Low Melting Mixtures

1.1 **Catalysts:** (+)-Tartaric acid, Dimethylolurea; 0.25 h, 70 °C

1.2 **Reagents:** Water

By: Gore, Sangram; et al

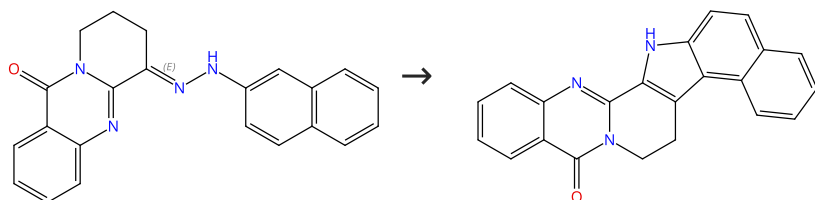
Organic Letters (2012), 14(17), 4568-4571.

Experimental Protocols

31-467-CAS-3679756 Steps: 1 Yield: 96% 1.1 Catalysts: 1 <i>H</i> -Imidazolium, 1,3-bis(3-sulfopropyl)-, 1,1,1-trifluoromethanesulfonate (1:1) Solvents: Water; 15 min, 100 °C	Novel SO₃H-functionalized ionic liquids catalyzed a simple, green and efficient procedure for Fischer indole synthesis in water under microwave irradiation By: Li, Bai Lin; et al Journal of Fluorine Chemistry (2012), 144, 45-50.
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Scheme 16 (1 Reaction)

Steps: 1 Yield: 98%



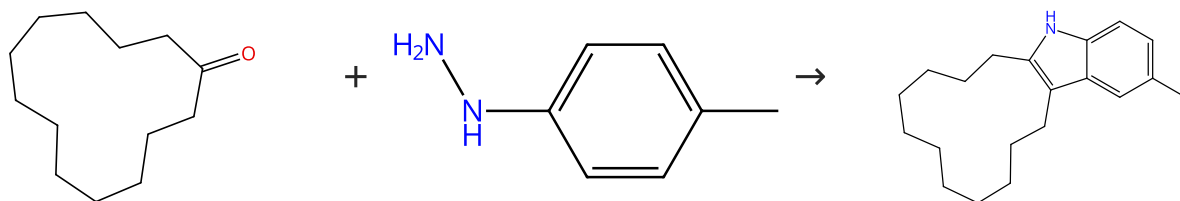
Double bond geometry shown

Supplier (1)

31-467-CAS-14343079 Steps: 1 Yield: 98% 1.1 45 min, 180 °C	Synthesis and biological properties of benzo- annulated rutaecarpines By: Hong, Young Hwan; et al Biological & Pharmaceutical Bulletin (2010), 33(10), 1704-1709.
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Scheme 17 (1 Reaction)

Steps: 1 Yield: 98%



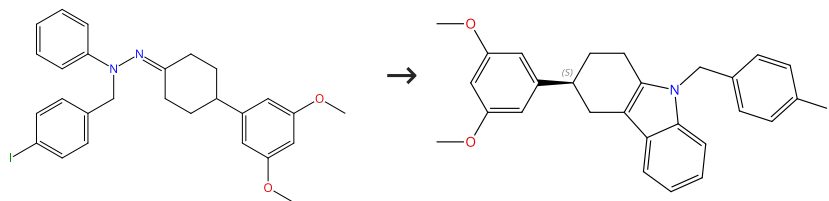
Suppliers (20)

Suppliers (54)

31-467-CAS-9219160 Steps: 1 Yield: 98% 1.1 Reagents: <i>p</i> -Toluenesulfonic acid; 1 min, 250 °C	Fischer indole synthesis in the absence of a solvent By: Matsumoto, Kiyoshi; et al Heterocyclic Communications (2003), 9(1), 9-12.
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Scheme 18 (1 Reaction)

Steps: 1 Yield: 98%

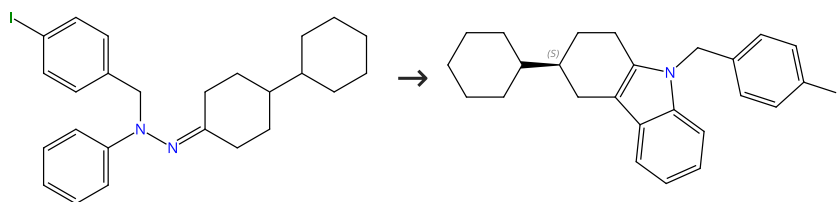


Absolute stereochemistry shown

31-467-CAS-3620517 Steps: 1 Yield: 98% 1.1 Catalysts: Diindeno[7,1- <i>de</i> :1',7'- <i>fg</i>][1,3,2]dioxaphosphocin, 3, 7-di-9-anthracenyl-10,11,12,13-tetrahydro-5-hydroxy-, 5-oxide, (11 <i>aR</i>)- Solvents: Benzene; 4 - 5 d, 30 °C Experimental Protocols	The Catalytic Asymmetric Fischer Indolization By: Mueller, Steffen; et al Journal of the American Chemical Society (2011), 133(46), 18534-18537.
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Scheme 19 (1 Reaction)

Steps: 1 Yield: 98%



Absolute stereochemistry shown

31-467-CAS-8146723

Steps: 1 Yield: 98%

The Catalytic Asymmetric Fischer Indolization

- 1.1 **Catalysts:** Diindeno[7,1-*de*:1',7'-*fg*][1,3,2]dioxaphosphocin, 3, 7-di-9-anthracenyl-10,11,12,13-tetrahydro-5-hydroxy-, 5-oxide, (11*aR*)-
Solvents: Benzene; 4 - 5 d, 30 °C

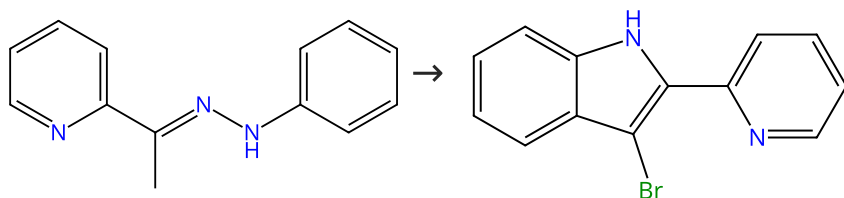
By: Mueller, Steffen; et al

Journal of the American Chemical Society (2011), 133(46), 18534-18537.

Experimental Protocols

Scheme 20 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (5)

Suppliers (9)

31-467-CAS-18795527

Steps: 1 Yield: 98%

One-Pot Reactions Involving the Fischer Indole Synthesis and Friedel-Crafts Reactions

- 1.1 **Catalysts:** Trifluoromethanesulfonic acid
Solvents: Chloroform; 24 h, 25 °C
- 1.2 **Reagents:** Hydrogen bromide
Solvents: Dimethyl sulfoxide; 3 h
- 1.3 **Reagents:** Water
- 1.4 **Reagents:** Sodium bicarbonate
Solvents: Water; pH 7 - 8

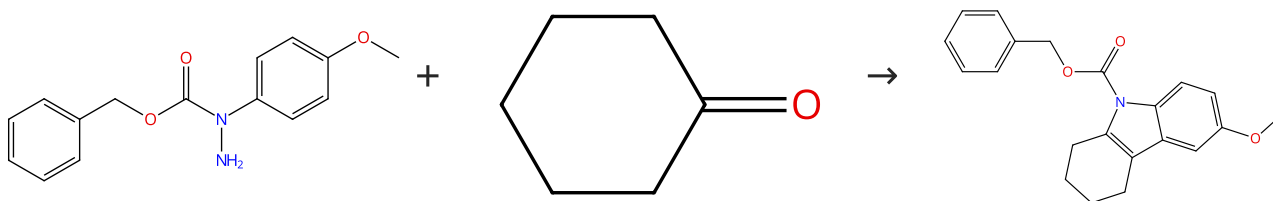
By: Vuong, Hien; et al

Topics in Catalysis (2018), 61(7-8), 685-688.

Experimental Protocols

Scheme 21 (1 Reaction)

Steps: 1 Yield: 98%



Supplier (1)

Suppliers (102)

Supplier (1)

31-467-CAS-14988313

Steps: 1 Yield: 98%

Aryl Hydrazide beyond as Surrogate of Aryl Hydrazine in the Fischer Indolization: The Synthesis of N-Cbz-indoles, N-Cbz-carbazoles, and N,N'-Bis-Cbz-pyrrolo[2,3-*f*]indoles

- 1.1 **Catalysts:** Sulfuric acid
Solvents: Methanol; 2 h, 50 °C

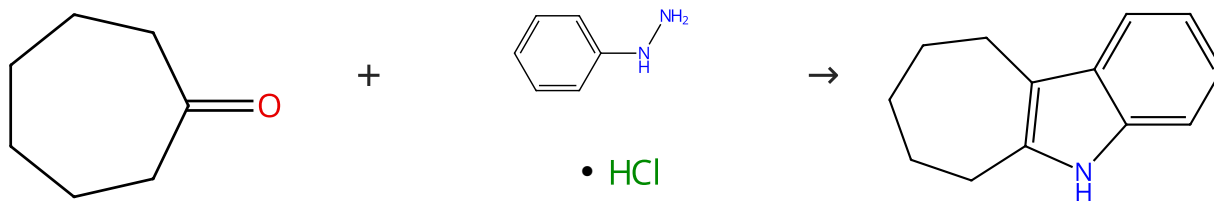
By: Park, In-Keol; et al

Organic Letters (2009), 11(23), 5454-5456.

Experimental Protocols

Scheme 22 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (65)

Suppliers (64)

Suppliers (53)

31-467-CAS-956829

Steps: 1 Yield: 98%

Fischer Indole Synthesis in Low Melting Mixtures

1.1 Catalysts: (+)-Tartaric acid, Dimethylolurea; 1 h, 70 °C

By: Gore, Sangram; et al

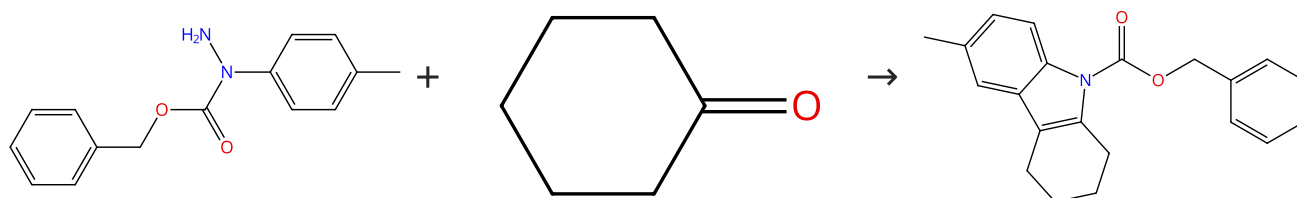
1.2 Reagents: Water

Organic Letters (2012), 14(17), 4568-4571.

Experimental Protocols

Scheme 23 (1 Reaction)

Steps: 1 Yield: 98%



Supplier (1)

Suppliers (102)

Supplier (1)

31-467-CAS-1516115

Steps: 1 Yield: 98%

Aryl Hydrazide beyond as Surrogate of Aryl Hydrazine in the Fischer Indolization: The Synthesis of N-Cbz-indoles, N-Cbz-carbazoles, and N,N'-Bis-Cbz-pyrrolo[2,3-f]indoles

1.1 Catalysts: Sulfuric acid
Solvents: Methanol; 2 h, 50 °C

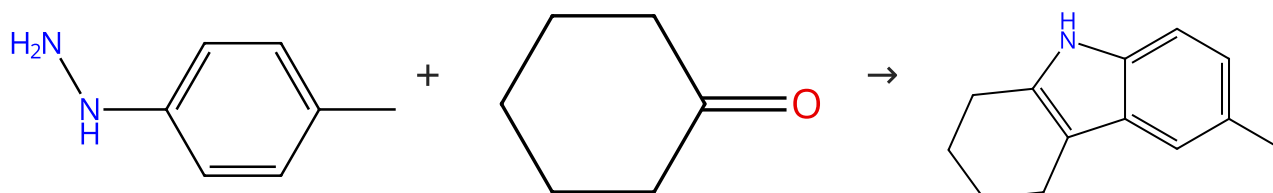
By: Park, In-Keol; et al

Organic Letters (2009), 11(23), 5454-5456.

Experimental Protocols

Scheme 24 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (54)

Suppliers (102)

Suppliers (36)

31-467-CAS-18139391

Steps: 1 Yield: 98%

Improved synthesis of 1,2,3,4-tetrahydrocarbazoles under ultrasonic irradiation

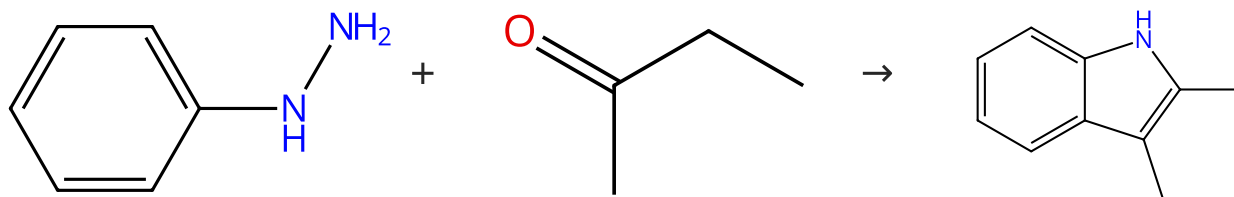
1.1 Catalysts: Acetic acid, Trifluoroacetic acid
Solvents: Ethanol; 18 min, 50 °C

By: Surendiran, T.; et al

Organic Chemistry: An Indian Journal (2008), 4(9-11), 478-481.

Scheme 25 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (64)

Suppliers (116)

Suppliers (73)

31-467-CAS-6908572

Steps: 1 Yield: 98%

Synthesis of substituted indoles using continuous flow micro reactors

1.1 Solvents: Ethanol; 10 min, rt

1.2 -

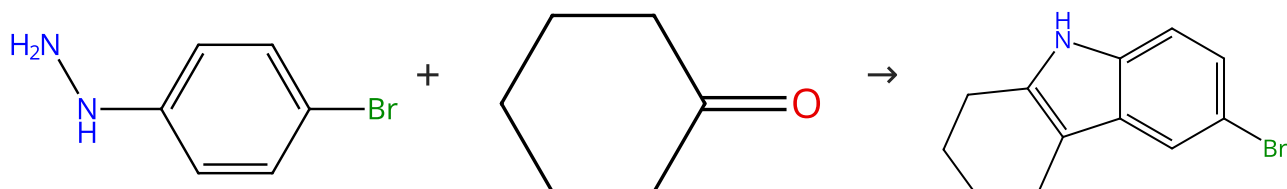
By: Wahab, Ben; et al

Experimental Protocols

Tetrahedron (2010), 66(21), 3861-3865.

Scheme 26 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (63)

Suppliers (102)

Suppliers (57)

31-467-CAS-18154017

Steps: 1 Yield: 98%

A microwave-assisted, propylphosphonic anhydride (T3P) mediated one-pot Fischer indole synthesis1.1 Catalysts: Propylphosphonic anhydride
Solvents: Ethyl acetate; 10 min, 110 °C

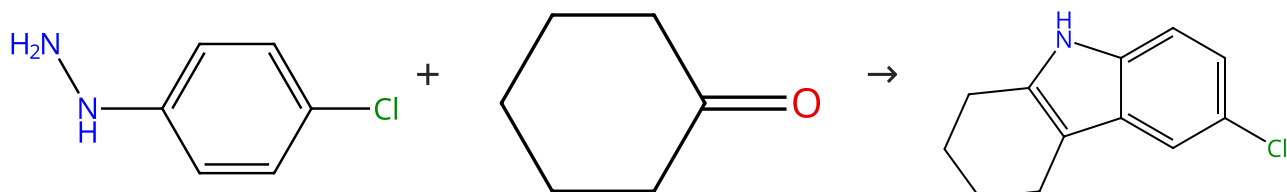
By: Desroses, Matthieu; et al

Experimental Protocols

Tetrahedron Letters (2011), 52(34), 4417-4420.

Scheme 27 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (65)

Suppliers (102)

Suppliers (54)

31-467-CAS-18129546

Steps: 1 Yield: 98%

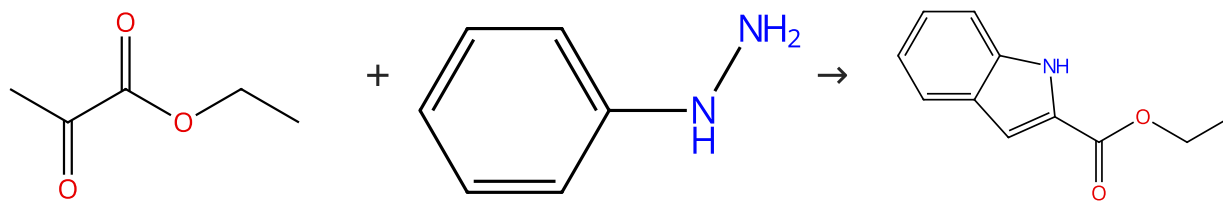
Improved synthesis of 1,2,3,4-tetrahydrocarbazoles under ultrasonic irradiation1.1 Catalysts: Acetic acid, Trifluoroacetic acid
Solvents: Ethanol; 16 min, 45 °C

By: Surendiran, T.; et al

Organic Chemistry: An Indian Journal (2008), 4(9-11), 478-481.

Scheme 28 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (99)

Suppliers (64)

Suppliers (90)

31-467-CAS-9038738

Steps: 1 Yield: 98%

Synthesis of substituted indoles using continuous flow micro reactors

1.1 Solvents: Ethanol; 10 min, rt

1.2 -

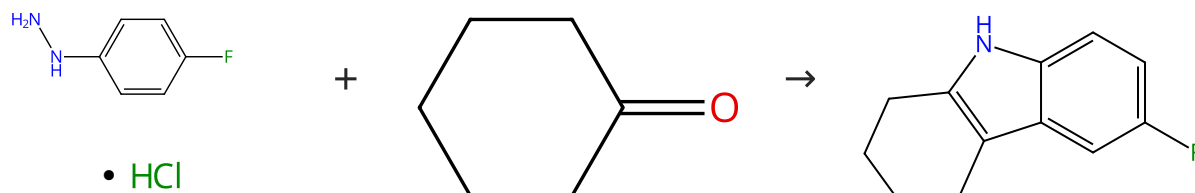
By: Wahab, Ben; et al

Experimental Protocols

Tetrahedron (2010), 66(21), 3861-3865.

Scheme 29 (1 Reaction)

Steps: 1 Yield: 98%



Suppliers (96)

Suppliers (102)

Suppliers (52)

31-467-CAS-11702194

Steps: 1 Yield: 98%

Mild, efficient Fischer indole synthesis using 2,4,6-trichloro-1,3,5-triazine (TCT)

1.1 Catalysts: Cyanuric chloride

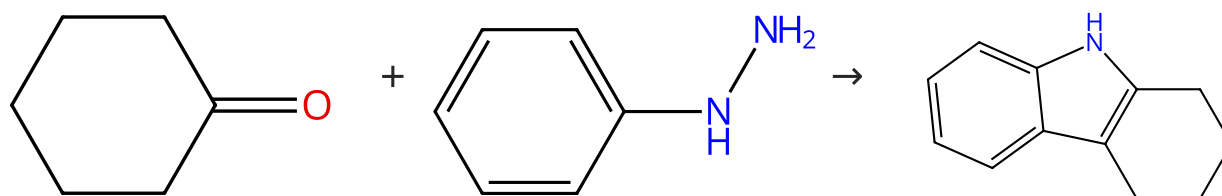
Solvents: Ethanol; 2.0 h, 80 °C; 80 °C → rt

By: Siddalingamurthy, Eranna; et al

Tetrahedron Letters (2013), 54(41), 5591-5596.

Scheme 30 (4 Reactions)

Steps: 1 Yield: 96-98%



Suppliers (102)

Suppliers (64)

Suppliers (70)

31-467-CAS-18152363

Steps: 1 Yield: 98%

A microwave-assisted, propylphosphonic anhydride (T3P) mediated one-pot Fischer indole synthesis

1.1 Catalysts: Propylphosphonic anhydride

Solvents: Ethyl acetate; 5 min, 100 °C

By: Desroses, Matthieu; et al

Experimental Protocols

Tetrahedron Letters (2011), 52(34), 4417-4420.

31-467-CAS-18135063

Steps: 1 Yield: 98%

Improved synthesis of 1,2,3,4-tetrahydrocarbazoles under ultrasonic irradiation

1.1 Catalysts: Acetic acid, Trifluoroacetic acid

Solvents: Ethanol; 14 min, 50 °C

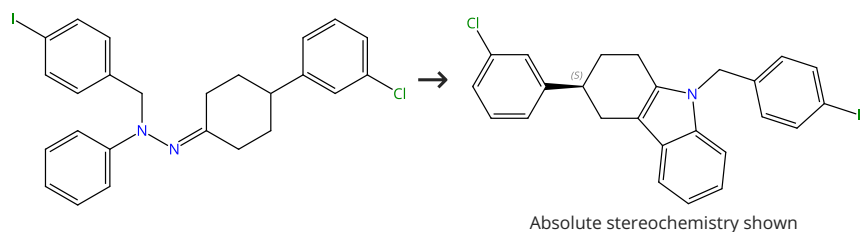
By: Surendiran, T.; et al

Organic Chemistry: An Indian Journal (2008), 4(9-11), 478-481.

31-467-CAS-18027945 Steps: 1 Yield: 97% 1.1 Catalysts: Acetic acid Solvents: Ethanol; 0.5 h, reflux 1.2 Catalysts: Zinc chloride Solvents: Triethylene glycol; 5 min, 100 °C Experimental Protocols	A New Approach to Difficult Fischer Synthesis: The Use of Zinc Chloride Catalyst in Triethylene Glycol under Controlled Microwave Irradiation By: Lipinska, Teodozja M.; et al Organic Letters (2006), 8(3), 367-370.
31-467-CAS-18079166 Steps: 1 Yield: 96% 1.1 Catalysts: Montmorillonite ((Al _{1.33-1.67} Mg _{0.33-0.67})(Ca ₀₋₁ Na ₀₋₁) _{0.33} Si ₄ (OH) ₂ O ₁₀ ·xH ₂ O) Solvents: Methanol; rt; rt → 60 °C; 5 h, 60 °C; 60 °C → rt Experimental Protocols	Facile clay-induced Fischer indole synthesis: A new approach to synthesis of 1,2,3,4-tetrahydrocarbazole and indoles By: Dhakshinamoorthy, Amarajothi; et al Applied Catalysis, A: General (2005), 292, 305-311.

Scheme 31 (1 Reaction)

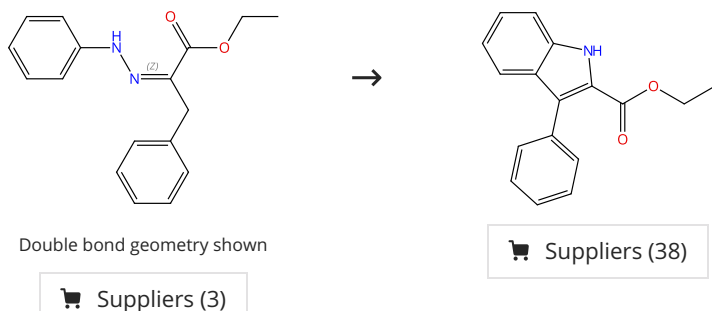
Steps: 1 Yield: 97%



31-467-CAS-7873788 Steps: 1 Yield: 97% 1.1 Catalysts: Diindeno[7,1- <i>de</i> :1',7'- <i>fg</i>][1,3,2]dioxaphosphocin, 3, 7-di-9-anthracenyl-10,11,12,13-tetrahydro-5-hydroxy-, 5-oxide, (11a <i>R</i>)- Solvents: Benzene; 4 - 5 d, 30 °C Experimental Protocols	The Catalytic Asymmetric Fischer Indolization By: Mueller, Steffen; et al Journal of the American Chemical Society (2011), 133(46), 18534-18537.
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Scheme 32 (1 Reaction)

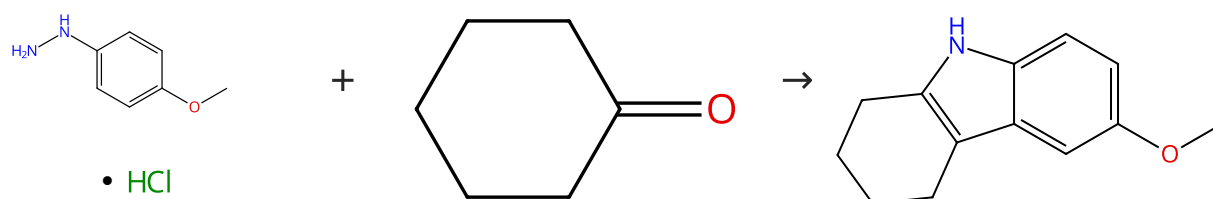
Steps: 1 Yield: 97%



31-467-CAS-3564669 Steps: 1 Yield: 97% 1.1 Reagents: Thionyl chloride Solvents: Ethanol; 0 °C; 40 min, 80 °C Experimental Protocols	Novel method for synthesis of aryl hydrazones from α-diazo esters: scope and limitations of nucleophiles By: Yasui, Eiko; et al Tetrahedron (2009), 65(2), 461-468.
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Scheme 33 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (92)

Suppliers (102)

Suppliers (52)

31-467-CAS-9573272

Steps: 1 Yield: 97%

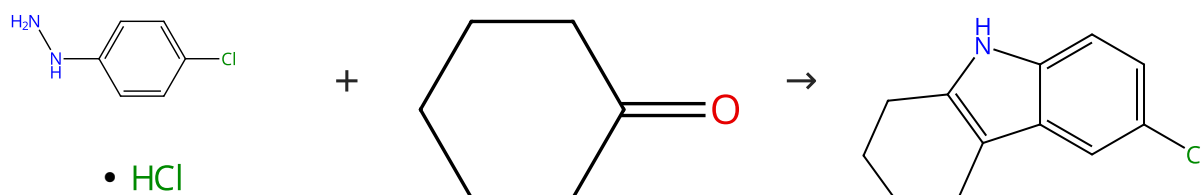
Mild, efficient Fischer indole synthesis using 2,4,6-trichloro-1,3,5-triazine (TCT)

- 1.1 **Catalysts:** Cyanuric chloride
Solvents: Ethanol; 2.0 h, 80 °C; 80 °C → rt

By: Siddalingamurthy, Eranna; et al
 Tetrahedron Letters (2013), 54(41), 5591-5596.

Scheme 34 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (92)

Suppliers (102)

Suppliers (54)

31-467-CAS-432003

Steps: 1 Yield: 97%

Fischer Indole Synthesis in Low Melting Mixtures

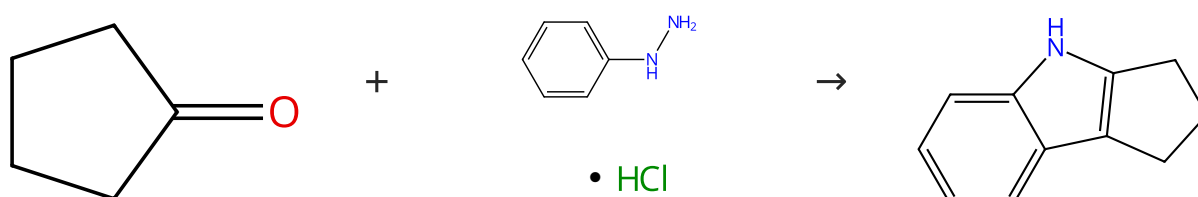
- 1.1 **Catalysts:** (+)-Tartaric acid, Dimethylolurea; 0.5 h, 70 °C
 1.2 **Reagents:** Water

By: Gore, Sangram; et al
 Organic Letters (2012), 14(17), 4568-4571.

Experimental Protocols

Scheme 35 (1 Reaction)

Steps: 1 Yield: 97%



Suppliers (78)

Suppliers (64)

Suppliers (63)

31-467-CAS-7320813

Steps: 1 Yield: 97%

Fischer Indole Synthesis in Low Melting Mixtures

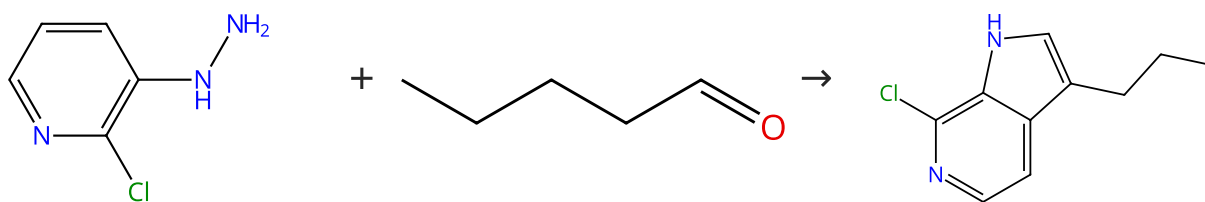
- 1.1 **Catalysts:** (+)-Tartaric acid, Dimethylolurea; 0.5 h, 70 °C
 1.2 **Reagents:** Water

By: Gore, Sangram; et al
 Organic Letters (2012), 14(17), 4568-4571.

Experimental Protocols

Scheme 36 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (51)

Suppliers (57)

Suppliers (3)

31-467-CAS-153917

Steps: 1 Yield: 96%

Extending the Scope of the Aza-Fischer Synthesis of 4- and 6-Azaindoles

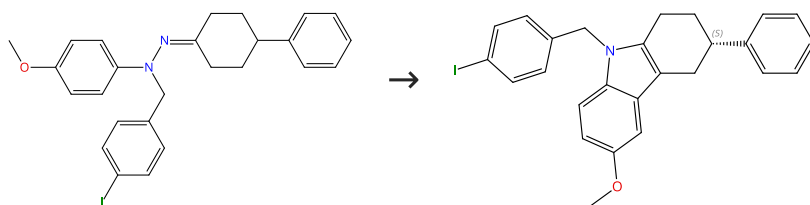
- 1.1 **Reagents:** Sulfuric acid
Solvents: Water; 1 min, rt; 20 min, 160 °C

By: Thomae, David; et al

European Journal of Organic Chemistry (2013), 2013(16), 3328-3336.

Scheme 37 (1 Reaction)

Steps: 1 Yield: 96%



Absolute stereochemistry shown

31-467-CAS-15199990

Steps: 1 Yield: 96%

The Catalytic Asymmetric Fischer Indolization

- 1.1 **Catalysts:** Diindeno[7,1-*de*:1',7'-*fg*][1,3,2]dioxaphosphocin, 3, 7-di-9-anthracenyl-10,11,12,13-tetrahydro-5-hydroxy-, 5-oxide, (11*aR*)-
Solvents: Benzene; 4 - 5 d, 30 °C

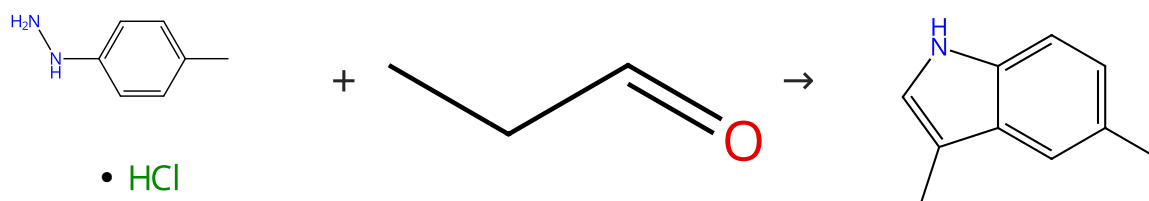
By: Mueller, Steffen; et al

Journal of the American Chemical Society (2011), 133(46), 18534-18537.

Experimental Protocols

Scheme 38 (1 Reaction)

Steps: 1 Yield: 96%



• HCl

Suppliers (84)

Suppliers (75)

Suppliers (18)

31-467-CAS-21036636

Steps: 1 Yield: 96%

Novel and Efficient Heterogeneous 4-Methylbenzenesulfonic Acid-Based Ionic Liquid Supported on Silica Gel for Greener Fischer Indole Synthesis

- 1.1 **Catalysts:** 1*H*-imidazolium, 1-(4-sulfobutyl)-3-[3-(trihydroxysilyl)propyl]-, 4-methylbenzenesulfonate (1:1) (silica supported)
Solvents: Ethanol; 4 h, rt

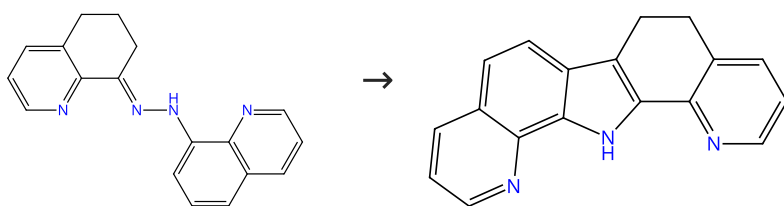
By: Hu, Yu-Lin; et al

Catalysis Letters (2016), 146(5), 968-976.

Experimental Protocols

Scheme 39 (2 Reactions)

Steps: 1 Yield: 96%



Supplier (1)

Supplier (1)

31-467-CAS-12049481

Steps: 1 Yield: 96%

No Data Available

Design of receptors for urea derivatives based on the pyrido [3,2-g]indole subunit

By: Hegde, Vidyadhar; et al

Journal of the American Chemical Society (1993), 115(3), 872-8.

31-467-CAS-1375473

Steps: 1 Yield: 96%

No Data Available

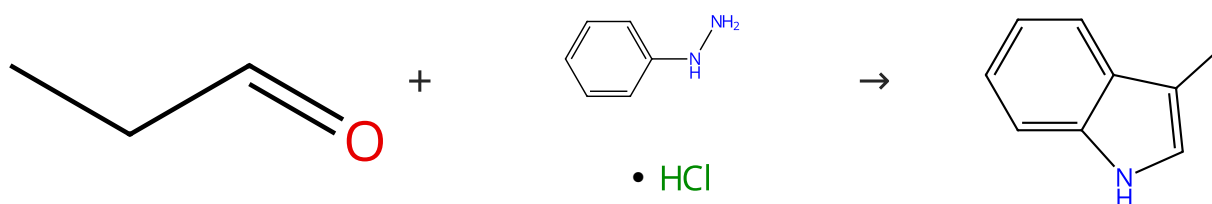
Fischer route to pyrido[3,2-g]indoles. A novel receptor for urea derivatives

By: Hegde, Vidyadhar; et al

Journal of the American Chemical Society (1990), 112(11), 4549-50.

Scheme 40 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (75)

Suppliers (64)

Suppliers (105)

31-467-CAS-21348585

Steps: 1 Yield: 96%

1.1 **Catalysts:** 1*H*-imidazolium, 1-(4-sulfobutyl)-3-[3-(trihydroxysilyl)propyl]-, 4-methylbenzenesulfonate (1:1) (silica supported)**Solvents:** Ethanol; 4 h, rt

Experimental Protocols

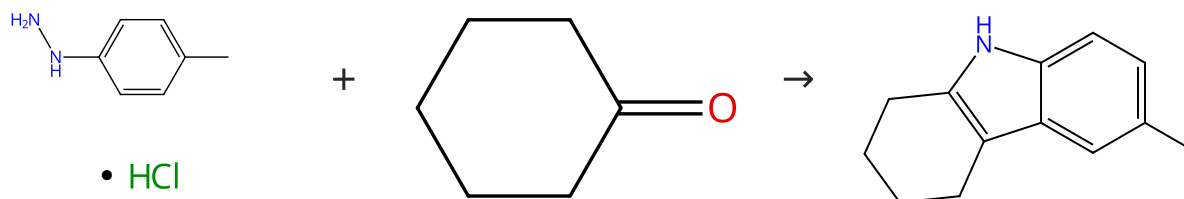
Novel and Efficient Heterogeneous 4-Methylbenzenesulfonic Acid-Based Ionic Liquid Supported on Silica Gel for Greener Fischer Indole Synthesis

By: Hu, Yu-Lin; et al

Catalysis Letters (2016), 146(5), 968-976.

Scheme 41 (1 Reaction)

Steps: 1 Yield: 96%



Suppliers (84)

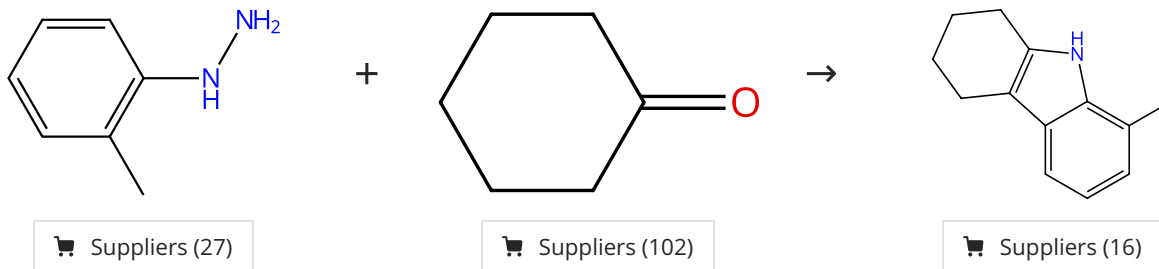
Suppliers (102)

Suppliers (36)

31-467-CAS-4746959	Steps: 1 Yield: 96%	Fischer Indole Synthesis in Low Melting Mixtures
1.1 Catalysts: (+)-Tartaric acid, Dimethylolurea; 0.25 h, 70 °C		By: Gore, Sangram; et al
1.2 Reagents: Water		Organic Letters (2012), 14(17), 4568-4571.
Experimental Protocols		

Scheme 42 (1 Reaction)

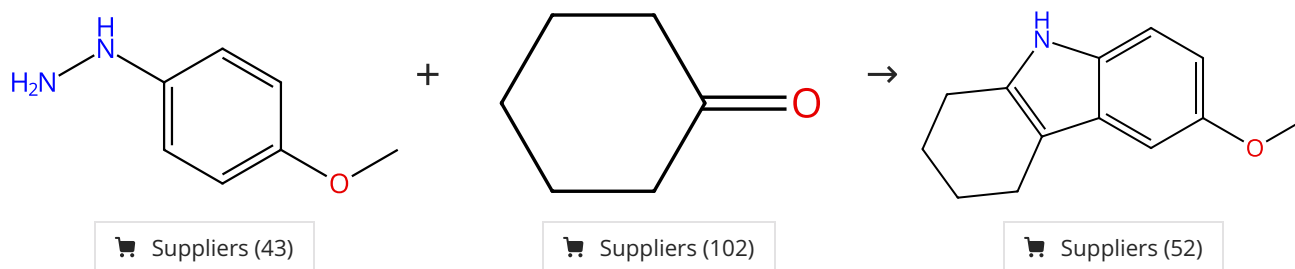
Steps: 1 Yield: 96%



31-467-CAS-18190213	Steps: 1 Yield: 96%	Fischer indole synthesis in Bronsted acidic ionic liquids: a green, mild, and regiospecific reaction system
1.1 Catalysts: 1-Butyl-3-methylimidazolium hydrogen sulfate Solvents: 1-Butyl-3-methylimidazolium hydrogen sulfate; 0.5 h, 70 °C; 70 °C → rt		By: Xu, Dan-Qian; et al
1.2 Reagents: Hydrochloric acid Solvents: Water; neutralized		European Journal of Organic Chemistry (2007), (6), 1007-1012.
Experimental Protocols		

Scheme 43 (1 Reaction)

Steps: 1 Yield: 96%



31-467-CAS-18152360	Steps: 1 Yield: 96%	Improved synthesis of 1,2,3,4-tetrahydrocarbazoles under ultrasonic irradiation
1.1 Catalysts: Acetic acid, Trifluoroacetic acid Solvents: Ethanol; 35 min, 50 °C		By: Surendiran, T.; et al
		Organic Chemistry: An Indian Journal (2008), 4(9-11), 478-481.